Amiya Islam

Miami, FL (open to relocation) | US Citizen | iamiya@vt.edu | linkedin.com/in/amiya-islam | amiyaislam.com

SKILLS

AI & Agents: Multi-agent orchestration, AI agents, LLM routing, tool/function calling, RAG, vector search (Pinecone), conversation memory, prompt engineering; OpenAI, Anthropic Claude, Gemini, Grok, Perplexity APIs; Composio; Claude Code, Codex, Gemini CLI, Warp

Cloud: AWS (Lambda, API Gateway, DynamoDB, S3, EC2, ECS, EventBridge, SQS, IAM, Secrets Manager, SAM, Amplify), Terraform, CloudFormation

Programming & Scripting: Python, Bash, JavaScript/TypeScript, Swift, HTML, CSS

Frameworks & Tools: Next.js, React, Node.js, Playwright, Auth0, Stripe, Chargebee, Twilio

CI/CD & Automation: Jenkins, Git, GitHub Actions, n8n

Containers, Monitoring & Data: Docker, Kubernetes, Prometheus, Grafana, CloudWatch, PostgreSQL, DynamoDB

Other: Linux, Networking & Security, Project Management, Jira

EXPERIENCE

Founding Engineer

February 2025 – Present

SWA AI (swa-ai.com)

Remote

Early-stage AI startup founded by former Nike executives

AI Agents & Platform

- **AI agent platform:** Built a production multi-agent AI platform on an open-source agent framework, used in Slack, Microsoft Teams, SMS, web, and mobile: AI agents work on their own with their own tools or coordinate with each other across files, images, video, deep research, web browsing, and scheduling, powered by OpenAI, Anthropic Claude, Gemini, Grok, Perplexity, and DeepSeek, with RAG (Pinecone vector search) and long-term conversation memory
- **Scale & ownership:** Primary author of 20+ backend services; shipped 1,000+ code changes to production across 35 codebases in 5 months
- **No-code agent builder:** Built a no-code agent builder where customers create custom AI agents by choosing instructions, model, knowledge documents, and per-app tool permissions; agents can be shared with the whole company or specific teams, or created as personal agents directly inside Slack, Microsoft Teams, or Swa Solo
- **Tool calling:** Launched a self-serve marketplace enabling customers to connect 121 business apps to their AI agents in one click; implemented lazy tool loading so agents only load the tools a request needs, keeping responses fast, cheap, and accurate
- **Autonomous agents:** Built the automation scheduler end to end on AWS serverless (Lambda, DynamoDB, EventBridge, SQS), running the full multi-agent system unattended on a schedule and delivering results to users; later unified scheduling across all messaging channels onto one system with automatic retry and failure alerts
- **Customer discovery & AI onboarding:** Worked directly with customers to understand their pain points and built a catalog of ready-to-use AI agents for their industries and teams, so agents matched real customer needs before launch; built AI-powered onboarding that reads a new customer's website, drafts their company profile, and recommends the right agents and models from that catalog, reducing manual setup for new accounts; also led a gamified learning and onboarding feature with progress tracking, streaks, badges, and in-chat reminders, rolled out gradually behind feature flags

Enterprise Controls, Billing & Data

- **AI governance:** Built enterprise controls for AI usage: monthly spend limits per person and per team enforced on every message, team management, team-only agents, model access controls at company, workspace, team, and user level, and limit-raise requests that admins approve or decline
- **Analytics:** Built a usage analytics service tracking AI usage and cost per user, team, and model across Slack, SMS, and Teams, powering billing reports and spend dashboards from millions of usage records
- **Billing:** Built the company's billing system end to end and led its migration from Chargebee to Stripe: selected the new provider, migrated all existing customer subscriptions, redesigned pricing, and built usage-based overage billing with spending caps; made billing safe against duplicate and out-of-order events, backed by 300+ tests
- **Identity & access:** Built the company's login and identity system on Auth0, consolidating separate login flows across 4 web apps into one secure, branded sign-in page with Google, Slack, and Apple sign-in; strengthened security checks across 8 production services and prepared enterprise SSO onboarding; tightened access controls across approximately 10 services so users can only reach their own organization's data, with a separate staff-only admin path and verified service-to-service communication
- **Data:** Led a database schema redesign across 9+ services to support multi-tenant customer organizations, reducing database costs and improving query performance; migrated live production data with zero downtime

Products & Channels

- **Product launch:** Built and launched Swa Solo, a standalone AI chat product, in approximately one week; owned both backend and Next.js frontend and later added paid plan upgrades via Stripe, marketing conversion tracking, team invites, voice input, and real-time streaming responses
- **Web portals:** Architected and built three web portals as primary author: customer admin dashboard, internal admin tool, and a resale partner portal; each includes billing, role-based permissions, task scheduling, AI agent management, live usage analytics, and automated testing
- **Messaging channels:** Built the SMS/WhatsApp assistant end to end on Twilio, matching the Slack and Teams versions with carrier compliance for high-volume texting, media and file delivery, conversation memory, and scheduling; also added a Telegram channel
- **Partner channel:** Co-built a partner reseller program covering wholesale pricing, revenue sharing, and automated invoicing, opening a new sales channel for the company

Leadership & Operations

- **Leadership:** Manage 2 engineers in India across time zones, covering task assignment, onboarding, code review, and input on performance, pay, and retention; collaborate daily with the CEO, CTO, and COO to turn business goals into shipped features
- **Reliability:** Set up a separate pre-production cloud environment fully defined in infrastructure-as-code; own production reliability including incident response, security audits, and dependency updates
- **AI code review:** Helped design the company's AI pull-request review agent after my shipping volume outpaced human review capacity; it reviews every PR across 30+ repos with parallel specialist AI reviewers (security, architecture, reliability, tests), verifies findings before posting to keep false positives low, and auto-merges only low-risk, CI-passing changes to test branches
- **AI adoption:** Lead AI tool evaluation company-wide, rolling out AI development tools (Claude Code, GitHub Copilot, Gemini CLI, Warp) to engineering and non-engineering teams; built the first company website with a demo-booking funnel and AI cost calculator; led the company's participation in the NVIDIA Inception program

Project Engineer May 2022 – August 2022
DPR Construction *Tampa, FL*

- Helped lead and coordinate the design-build construction of a $100 million Municipal Center for the City of Tampa
- Managed a diverse team and coordinated RFIs, meetings, and document management to meet stringent project milestones
- Ensured quality control and schedule adherence using PlanGrid, Bluebeam, CMiC, and Navisworks

Supervisor May 2019 – August 2019
Virginia Tech Division of Student Affairs *Blacksburg, VA*

- Directed and trained a team of 30 in a campus operations role; tripled efficiency by revamping operational procedures
- Owned scheduling, performance management, and resource coordination across internal departments

EDUCATION

Virginia Tech Blacksburg, VA
M.S. in Information Technology (enrolled, paused to work full-time) 2025 –
Coursework: Programming, Networking, System Design, Cybersecurity, AI & ML

Virginia Tech 2023
Certificate in DevOps Engineering. GPA: 4.0. Coursework: Linux, Networking, Python, AWS, CI/CD, Docker, Jenkins

Virginia Tech 2023
B.S. in Construction Engineering and Management. GPA: 3.2. Coursework: Project Management, Estimating, Scheduling, Structural Analysis

CERTIFICATIONS

- AWS Certified DevOps Engineer, Professional (2024); AWS Certified Security, Specialty (2024)
- AWS Certified Solutions Architect, Developer, and SysOps Administrator, Associate (2023–2024)
- CKA: Certified Kubernetes Administrator (2024); LFCS: Linux Foundation Certified Systems Administrator (2024)
- CompTIA Security+ (2025); HashiCorp Certified: Terraform Associate (2023); Chainguard Painless Vulnerability Management (2024)

ACTIVITIES & HOBBIES

Volunteer, Warm Hearth Village January 2022 – Present

- Provide fitness and cooking classes, social interaction, and administrative support for senior citizens

Hobbies: Powerlifting, trading, health, fitness and wellness, the outdoors and nature, and recreational sports